Ergonomic assessment

VIDEO	19832490-hd_1920_1080_25fps (1).mp4
ANALYZED	08 Oct 2026 · 13:48 UTC
DURATION	5.8 s
ASSESSMENT	cb03a719-ecc6-4642-9420-36d891f7bb27

01 OVERALL ASSESSMENT

MODERATE

ERGONOMIC RISK

RULA	04 / 07	Moderate
REBA	04 / 15	Medium

Posture risk was moderate during the analyzed movement. The main opportunities are to reduce forward trunk bending and reduce deep knee flexion.

LIMITED MEASUREMENT RELIABILITY

The camera was close to side-on in only 3% of measured frames. Trunk lean in particular may be larger or smaller than measured.

For a confident assessment, record a new video with the camera to the side of the worker and the whole body in view. This assessment and its results are kept as recorded.

02 PRIORITY ACTION

BACK / TRUNK

PROTECT THE BACK

Keep the back closer to a neutral position where practical. Reduce repeated forward bending by bringing the task closer or adjusting the work surface.

Forward trunk positioning was observed repeatedly during the assessment, beyond the upright range in 6 of 29 measured frames (peak 13°). The camera was not side-on for most of the video, so forward bending may be larger than measured.

Status: Attention

03 RECOMMENDED PRODUCT TO EVALUATE

EXTRIVE INNOVATIONS · PASSIVE BACK-SUPPORT EXOSUIT

Evaluate BackEX for this task

The assessment observed repeated forward trunk bending, beyond the upright range in 6 of 29 measured frames. Alongside improving task height, reach and work positioning, consider evaluating Extrive's BackEX support solution for tasks involving repeated or sustained bending. Review whether its intended support, fit and operating requirements match this task before introducing it.

WHY IT MAY HELP

BackEX is a passive back-support exosuit designed for repetitive lifting, forward bending and load-carrying tasks in industrial, warehousing, manufacturing and logistics work. It is a lightweight, fabric-based wearable with no batteries or motors. Elastic elements are tensioned as the wearer bends forward and return that stored energy as the trunk straightens, so part of the effort needed to control and reverse forward bending is shared by the suit instead of being carried by the lower-back muscles alone.

Relevant tasks: repeated or sustained forward bending; repetitive lifting and lowering; load carrying and material handling, such as warehouse picking, parcel loading and unloading, inventory movement, assembly-line lifting and fabrication work.

BEST RESULTS COME FROM COMBINING APPROACHES

- Adjust the workstation and work height where practical.
- Bring tools and materials closer to reduce unnecessary reaching and bending.
- Avoid prolonged or repeated awkward postures where possible.
- Evaluate BackEX fit, task suitability, comfort and worker acceptance before use.

BEFORE INTRODUCING IT

- Confirm the task involves the repeated or sustained forward bending BackEX is designed to support.
- Have a qualified person check fit, comfort and freedom of movement with the worker before regular use.
- Check that it can be worn with the clothing and protective equipment the task requires.
- Run a trial period and gather the worker's feedback on comfort and acceptance.

This report makes no claim about the size of any BackEX benefit. Evaluate whether it improves task comfort and work efficiency in your operating environment; Extrive can share its product evaluation material on request.

This video assessment observed posture only. It does not establish that this worker needs BackEX or that it will suit them, and BackEX does not replace correcting the work design. Have a qualified person assess fit and task suitability before introducing it.

Product information: Extrive BackEX brochure (Brochure_Extrive_BackEX.pdf); Extrive BackEx scientific validation report, section 2: the BackEx exosuit and its working principle (2025).

04 OTHER ACTIONS

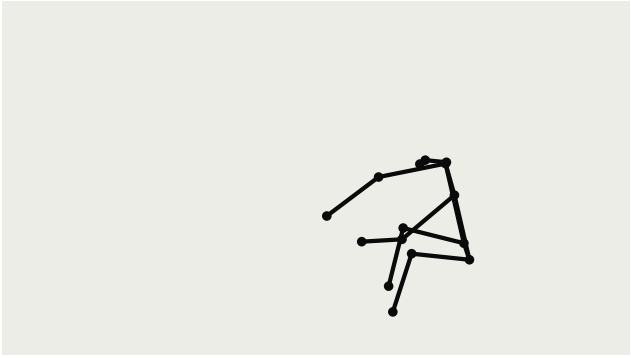
	ACTION	STATUS
01	**REDUCE DEEP KNEE BENDING** Where practical, raise or reposition the work area to reduce prolonged or repeated deep knee flexion. Deep knee bending was observed consistently during the assessment, in 27 of 27 measured frames (peak 107° of knee flexion).	Elevated
02	**KEEP WORK WITHIN A COMFORTABLE REACH** Position frequently used work within a comfortable reach to reduce prolonged arm elevation. Raised upper-arm positions were observed consistently during the assessment, in 29 of 29 measured frames (peak 55° from the body).	Elevated
03	**REDUCE NECK BENDING** Bring the task closer to eye level where practical and avoid prolonged downward viewing. Neck bending away from upright was observed repeatedly during the assessment, in 19 of 29 measured frames (peak 23°).	Elevated

05 MEASUREMENTS

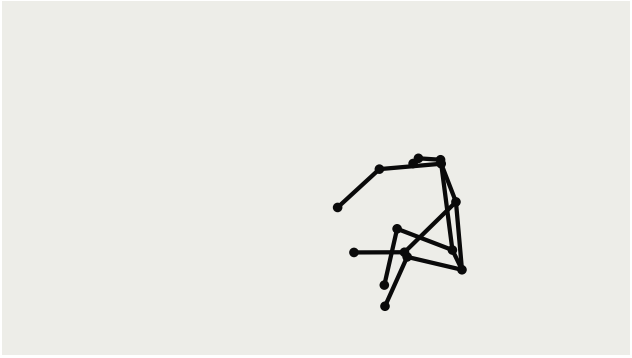
REGION	AVERAGE	PEAK	STATUS
Back / trunk	6.3°	13.1°	Attention
Upper arm	45.3°	55.3°	Elevated
Knee flexion	78.2°	107.2°	Elevated
Neck	12.4°	23.1°	Elevated

Average and peak joint angles across all analyzed frames in which each region was observable. Not measured means the region could not be observed in any scored frame. It is not evidence that the posture of that region was low risk.

06 POSTURE EVIDENCE



FRAME 27 · 5.2 s
PEAK TRUNK 13.1°



FRAME 01 · 0.0 s
PEAK REBA 04 / 15

Detected pose from the stored landmarks. The original video is not retained by the server.

07 CAMERA VIEW AND RELIABILITY

Joint angles are estimated from 2D pose landmarks. They are most reliable when the worker is clearly visible from a side view. Oblique, rear, frontal or partly hidden views can distort the estimated angles, especially trunk lean, and may understate or overstate a posture.

Measured frames filmed close to side-on	3%
Sampled frames with the worker detected	100%

LIMITED MEASUREMENT RELIABILITY

The camera was close to side-on in only 3% of measured frames. Trunk lean in particular may be larger or smaller than measured.

For a confident assessment, record a new video with the camera to the side of the worker and the whole body in view. This assessment and its results are kept as recorded.

A TECHNICAL DETAILS AND ASSESSMENT LIMITATIONS

RULA / REBA SCORES

RULA 04 / 07 · moderate risk. 29 frames scored. Frames by risk band: low 9, moderate 20.

REBA 04 / 15 · medium risk. 29 frames scored. Frames by risk band: low 9, medium 15, negligible 5.

Scores are the peak per-frame score reported by the VigilEx pipeline; band is the component's score within its range.

COMPONENT MEASUREMENTS AT THE PEAK-SCORE FRAME

The overall peak shown in Measurements is the highest observed value across analyzed frames. Component angles below are taken from the frame where the RULA or REBA score reached its peak, so they may differ.

RULA — component measurements at the peak-score frame score 04 / 07 · frame 1 · 0.0 s

COMPONENT	ANGLE	BAND
Upper arm	45.6°	3 / 4
Lower arm (elbow)	44.1°	2 / 2
Wrist	Not observable	—
Wrist twist	Not observable	—
Neck	20.1°	3 / 3
Trunk	5.1°	1 / 4
Legs	79.0°	2 / 2

REBA — component measurements at the peak-score frame score 04 / 15 · frame 1 · 0.0 s

COMPONENT	ANGLE	BAND
Upper arm	45.6°	3 / 4
Lower arm (elbow)	44.1°	2 / 2
Wrist	Not observable	—
Neck	20.1°	2 / 2
Trunk	5.1°	1 / 4
Legs	79.0°	3 / 3

METHODOLOGY

VigilEx detects the worker's pose in sampled video frames (YOLO11n-Pose) and follows the same worker through the video (ByteTrack). Joint angles are measured from the detected 2D landmarks.

Frames were sampled at 5 per second (29 frames sampled).

RULA and REBA are scored per sampled frame from those angles; this report gives the peak score of each and how the scored frames were distributed across risk bands.

Measurements give the average and highest value of each angle across all analyzed frames. Component tables give the angles at the single frame where each method reached its peak score.

LIMITATIONS

Wrist posture, wrist twist, force or load, coupling and activity cannot be observed from the video and are scored at their lowest-risk values. This is not evidence that they were low risk.

Joint angles are estimated from 2D pose landmarks. They are most reliable when the worker is clearly visible from a side view. Oblique, rear, frontal or partly hidden views can distort the estimated angles, especially trunk lean, and may understate or overstate a posture.

Limited measurement reliability: The camera was close to side-on in only 3% of measured frames. Trunk lean in particular may be larger or smaller than measured.

This report describes observed posture. It does not assess injury, pain, fatigue or health, and findings should be reviewed by a qualified person.

Recommended interventions are general workplace and posture suggestions based on the observed posture, not medical advice.

Product suggestions are for evaluation only. Product descriptions come from Extrive product material; the posture scores in this report are not evidence of a product's effectiveness.

Methodology version: vigilex-rula-reba-v1